MILITARY
SPECIFICATION

MIL-R-55217(EL)
27 February 1963

RECORDER-PROCESSOR-VIEWER, RADAR MAPPING RO-166( )/UP

1. SCOPE

1.1 Scope.- This specification covers one type of recorder, processor and viewer designated as Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP. (See 6.3.)

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference control requirements, Aircraft Equipment
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper-Overlaid

FSC 5841

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme Temperature Cycle
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

DRAWINGS

SIGNAL CORPS

SC-A-362100	Requirements for Preparation of Packaging and Packing Illustrations and Bills of Material

(Copies of specification, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP is an airborne equipment designed to record on film the intensity modulated single line scan information displayed on two 5-inch cathode ray tubes, process, and present the finished pictures for immediate viewing. The optical system shall be capable of combining the two scans so that they are side by side on 9 1/2 inch wide film.

3.2 Preproduction samples.- The contractor shall furnish two (2) preproduction samples of Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP for approval, if required by the invitation for bids and contract. (See 4.3.)

3.3 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.3.1 Physical construction.- Physical construction of the equipment shall conform to the model.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:14Z
Check the source to verify that this is the current version before use.

3.3.2 Features.- The equipment shall incorporate all features of the model.

3.3.3 Performance.- The contractor shall adjust the model for optimum performance and then shall measure its performance characteristics. Performance of the equipment shall conform to those measurements, except that, in case of conflict between specified performance characteristics and the performance of the model, specified performance characteristics shall govern.

3.4 Parts, materials, and processes.- In addition to the requirements of this specification, the requirements of Specification MIL-E-5400, including the selection requirements therein, shall apply. The class, per MIL-E-5400, shall be Class 1 except the altitude shall be 15,000 feet and the temperature range shall be +5°C to +55°C.

3.5 Camera.- The camera portion of the equipment shall optically transfer the traces from two cathode ray tubes onto the sensitized material (film). (See 3.6.2.) Each cathode ray tube shall utilize 4 inches of film width.

3.5.1 Lenses.- The lenses, which transfer the images from the cathode ray tubes to the film, shall have an overall magnification of 1:1 ± 2 percent and shall cover a field at least 1 1/2 inches by 4 1/2 inches.

3.5.2 Data chamber.- The data chamber shall permit the recording of the following data, with enough clarity to be read with the magnifier (see 3.7.1), in a one-half inch width and two inch length along the edge of the film.

a. Drift angle (meter read-out)
b. Ground speed (meter read-out)
c. Time (clock)
d. Heading (drum type compass dial, syncho driven)
e. Range delay (three lights)
f. Range (two lights)
g. Write-in card
h. Locating mark

3.5.2.1 Lamps.- The operator shall expose the data on film by means of flash lamp controlled by a momentary lamp switch. The lamps shall be capable of 5000 operations without the necessity of a change in f stops.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.6 Processor.- The processor shall utilize a monobath processing technique. The monobath solution shall be Eastman Kodak Monobath MX-404A, or equal. The film processing shall be such as to preserve the radar information in normal room lighting for 24 hours minimum without further processing. (Permanent fixing shall be possible with commercially available fixers.) Normal room lighting is considered to be 50 ft-candles at table height (28 to 30 inches).

3.6.1 Film drive.- The film shall move through the equipment with uniform velocity. The film drive servo shall be a Norden-Ketay Model No. 105P2R Integrating Motor-Tachometer, or equal. When tested as specified in paragraph 4.7, the arithmetic mean of the readings shall fall within the range of .00498 to .0060 inches and the standard deviation computed in accordance wtih standard statistical methods (A.S.T.M. Manual), shall not exceed .0004.

3.6.2 Sensitized material (film).- The film used in the equipment shall be 9 1/2 wide Eastman Kodak SO-261 film, or equal.

3.6.2.1 Processing.- Film, which has been processed by the equipment and subsequently fixed and washed as specified (see 3.6), shall have a H and D curve shape similar to typical curves for SO-261 film from conventional processing. Good processing shall be maintained at any angle of static tilt less than 10° from the normal operating position of the equipment. (See 4.11.)

3.6.2.2 Fogging.- The fog level shall not exceed 0.2 for film speeds uo to 0.4 inches per minute; above 0.4 inches per minute, the fog level shall not exceed a density of 0.4. (See 4.8.)

3.6.2.3 Density.- A maximum density of 1.5 or higher shall result from a log exposure range of 2.0 when tested at +21°C or higher. When tested at +5°C, the maximum density shall be 1.2 or higher. (See 4.11.)

3.6.3 Exposure to view distance.- The distance between exposure and viewing shall not be more than 0.45 inches. Viewing area is defined as starting at the bottom of the illuminated glass.

3.6.4 Processing endurance.- The equipment shall be capable of holding enough solution for a minimum of 3 hours of operation. (See 4.12.)

3.6.5 Film storage.- The equipment shall be provided with cassettes capable of storing a film strip of not less than 50 feet in length. There is no leader or trailer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.5.1 Footage indicator.- A footage indicator shall be provided on each cassette to indicate the number of feet of film remaining in the cassette. The indicator shall never read more than the actual footage remaining but shall not read less by more than 20 percent.

3.6.5.2 Take-up spool.- The film take-up spool shall be identical to the type of spool on which SO-261 film is supplied.

3.6.6 Leakage, fluid (see 4.10).-

3.6.6.1 External.- The fluid shall not leak from the housing of the equipment when tilted at any angle up to 60° from the normal operating position. When tilted at 60° leakage through the cathode ray tube aperture shall not constitute a failure. (Sealing between the processor and radar indicator shall be effected by a light and fluid tight gasket.)

3.6.6.2 Internal.- There shall be no leakage within the processor housing when the processor is operated at a static tilt angle up to and including 30° from the normal operating position.

3.6.7 Pump.- The delivery rate of the pump shall be 2.4 ± 0.3 cubic centimeters per minute.

3.7 Viewer.- The viewer shall provide the capability of viewing details on the processed film in the density range between 0.15 and 2 under daylight conditions but not with sunlight directly on the viewing area.

3.7.1 Magnifier.- The magnifier shall have a magnification of 2 1/2 times.

3.7.2 Viewing hood.- A viewing hood shall be provided to facilitate viewing the details on film in high ambient light.

3.8 Transit case.- A transit case, in accordance with the model (see 3.3), shall be provided.

3.9 Weight.- The weight of the equipment, including solution and film, shall not exceed 51 pounds.

3.10 Service conditions.- The equipment, unless otherwise specified, shall meet the service conditions requirements for Class one (1) equipment of Specification MIL-E-5400, when tested in accordance with the requirements and test procedures of Specification MIL-T-5422.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.1 Exceptions to Specification MIL-E-5400.- The following exceptions to MIL-E-5400 shall apply:

3.10.1.1 Altitude-temperature.-

a. Equipment non-operating (fluid and film present).- Barometric pressure ranging from sea level to 50,000 feet and temperature from +5°C to +38°C.

b. Equipment operating (fluid and film present).- Barometric pressure ranging from sea level to 15,000 feet, the pressure remaining constant for long periods of time or varying at a rate as high as 0.5 inches of mercury per second and temperature from +5°C to +55°C.

3.10.1.2 Bounce and shock (no fluid present).- In addition to the shock specified in MIL-E-5400, the equipment, in its transit case, shall withstand the following:

a. Drop - a drop of 30 inches

b. Bounce - No damage when tested as specified in 4.14.2.1

3.11 Resolution.- The equipment shall be capable of resolving 12 1/2 or more lines per millimeter over the entire viewing area at maximum lens aperture. (See 4.13.)

3.12 Standby operation.- The equipment shall be capable of processing film after a standby period of at least 4 hours between operating periods without disassembly or cleaning. Processing shall be as specified in 3.6.2.1 twenty (20) minutes after re-start. (See 4.15)

3.13 Electrical.- The equipment shall be capable of operating from the following power sources:

Voltage	Maximum power dissipation	Input regulation
115 V AC, 400 cps	30 watts	$^{+10}_{-0}$ percent
28 V DC	300 watts	as specified in MIL-STD-704
+250 V DC		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.13.1 Breakdown voltage, insulation.- After 10-second application of 500 volts breakdown voltage, there shall be no visible deterioration and the equipment shall meet specified performance requirements. (See 4.6.1.)

3.13.2 Insulation resistance.- Insulation resistance shall exceed 100 megohms. (See 4.6.2.)

3.14 Interference suppression.- Interference control requirements shall be in accordance with Specification MIL-I-6181.

3.15 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9.) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.16 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.17 Workmanship.- The equipments shall be manufactured and assembled in accordance with the applicable portions of Specification MIL-E-5400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5.)

(2) Acceptance inspection of preparation for delivery. (See 4.18.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B (see tables II and III, respectively).

Table I.- Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Requirement paragraph	Inspection paragraph
Temperature	3.10	4.14.1.2
Elevation	3.10	4.14.1.1
*Vibration	3.10	4.14
*Shock	3.10	4.14
Bounce	3.10	4.14.2.1
Shock, drop	3.10	4.14.2.2
*Humidity	3.10	4.14
*Salt spray	3.10	4.14
*Explosion proof	3.10	4.14
*Sand and dust	3.10	4.14
*Fungus	3.10	4.14
Suppression	3.14	4.16

* Requirements in accordance with MIL-E-5400 and Tested in accordance with MIL-T-5422.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.4 per MIL-E-5400
Marking	3.4 per MIL-E-5400

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II .- Group A inspection

Inspection	Req. para.	Insp. para.	AQL	
			Major	Minor
Visual and Mechanical	3.17	4.17	1.0%	4%
Electrical				
Breakdown voltage	3.13.1	4.6.1	1.0%	
Insulation resistance	3.13.2	4.6.2	1.0%	

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Film drive	3.6.1	4.7	6.5%
Fogging	3.6.2.2	4.8	for the
Leakage	3.6.6.1, 3.6.6.2	4.10	entire group
Operational	3.5.2, 3.6.2.1, 3.6.2.3, 3.6.3	4.11	
Interchangeability	3.15	4.9	6.5%

4.5.3 Group C inspection.- This inspection shall be as listed in table IV and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV - Group C inspection

Inspection	Reqt Para	Test Para
Processing Endurance	3.6.4	4.12
Resolution & Magnification	3.5.1 & 3.11	4.13
Standby operation	3.12	4.15

4.5.3.1 Sampling.- Two Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP shall be selected for Group C inspection, without regard to their quality, at the start of the contract, from the first units produced and one unit out of every twenty-five units produced thereafter. The number of samples shall not exceed two per month.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Electrical.-

4.6.1 Breakdown.- The equipment shall be tested for conformance to 3.13.1. During testing, disconnections shall be made, where necessary, to protect circuit elements. Specified performance after this test shall be as follows:

Paragraph	Measurement
4.11	Operation

4.6.2 Insulation resistance.- The equipment shall be tested to determine compliance with the requirement specified in 3.13.2.

4.7 Film drive.- With the equipment operating, expose 3.0 inches of slit images on the film at a film speed of 0.1 inch per minute using the equipment specified in 4.11.1(b). After the film has been processed, fixed and washed, measure the spacing between the leading edges of the slit images which have been created by the 3.0 second cycle timer (2 seconds off - 1 second exposure). The sample shall consist of at least 250 consecutive readings. The arithmetic mean of the 250 readings shall be determined and shall be as specified in paragraph 3.6.1.

4.8 Fogging.- Load the equipment with unexposed film (Eastman Kodak No. SO-261, or equal), under darkroom conditions, cover the recorder lenses with lens caps and close all access doors. Do not cover the rectangular CRT apertures. Seal the viewing window from external light. Set the illuminator switch, on the equipment, to bright and remove the data card, leaving the data card cover off. Operate the equipment, without any processing solution, while exposing the equipment to a minimum of 7000 ft-lamberts light source for 15 minutes. Following the 15 minutes exposure, remove the film, under darkroom conditions, and develop using Kodak D-19 developer, or equal, fix, wash and dry the film. Using a densitometer, check the maximum fog level of the film for compliance with the requirement specified in paragraph 3.6.2.2. The 7000 ft-lamberts light source shall be moved about the equipment at a distance of three feet from the equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

4.9 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

4.10 Leakage.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP shall be tested to determine compliance with the requirements specified in paragraphs 3.6.6 through 3.6.6.2 inclusive.

4.11 Operational.-

4.11.1 Test conditions.-

(a) Load the equipment with film which has been exposed with sets of step wedges consisting of one step wedge located one inch from each edge of film and one at center of film.

(b) Mount the equipment on a Illuminated Target Fixture (resolution and slit targets), Eastman Kodak, Part No. CB-71, or equal. Apply power to the fixture thru a 3 second, cycle timer switch.

(c) Connect the equipment to a Servo Drive Fixture having the following associated power supplies:

+250 V DC ± 10 V and 100 ma
-300 V DC ± 5 V and 20 ma

(d) Fill the developer tank of the equipment with 1 pint of filtered developer and leave the upper and lower front access doors open.

(e) Write the test date and the equipment serial number on the data chamber card and place card in the data chamber.

(f) Wind the clock in the data chamber and apply power to the equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

(g) Turn the motor switch on and adjust the Servo Drive Fixture to provide an average film velocity within the range of 0.1 to 0.12 inch per minute.

4.11.2 Test method.-

(a) Allow the equipment to warm-up for 15 minutes.

(b) During warm-up period, check all controls and switches for operation.

(c) Check pump delivery rate to determine compliance with requirement specified in 3.6.7.

(d) Set the cathode ray tube lens aperture at f/22, remove filters and close access doors.

(e) With the film advancing as adjusted in 4.11.1(g) process one set of density wedges on the film while making the following exposures:

(1) Make two or three data chamber exposures at approximately 1.5 inch intervals.

(2) Turn on the illuminated target fixture and timer switch and expose 3.0 inches of slit images which shall appear on either side of the developed step wedges.

(f) Following the processing of the wedges at 0.1 inch/min film speed, set film speed to 0.4 inch/min and process one set of density wedges on the film.

(g) Set film speed to 0.75 inch/min and process one set of density wedges on the film.

(h) Replace the processor roller with roller which is supplied as a running spare, under darkroom conditions.

(i) Continue to process the pre-exposed film at a film speed of 0.1 inch/min for a minimum of 1 hour. Re-filling of the solution tank is permissible to attain 1 hour operation.

(j) Remove, wash and dry the processed film.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(k) Examine the processed film to determine compliance with the following requirements:

3.5.2	Data chamber
3.6.2.1	Processing
3.6.2.3	Density

(l) The film remaining in the supply cassette shall be measured to determine compliance with the requirement specified in 3.6.5.1.

4.12 Processing endurance.- The equipment shall be set up for operation and operated for 3 hours to determine compliance with the requirement specified in paragraph 3.6.4.

4.13 Resolution and magnification.- Set the equipment up for operation as specified in paragraph 4.11.1(a) and (b). Set the lens apertures at f/4.5 and close the access doors. Seal the viewing window from external light. Make several different time exposures of the target reticle. Remove the exposed film in the darkroom and develop, fix, wash and dry the film. Examine the processed film for resolving power to determine compliance with paragraph 3.11. Check magnification be measuring across the outside edges of the two slit images on each side of the film and compute the magnification as follows:

$$\text{Magnification} = \frac{\text{Distance on film}}{\text{Distance on chart}}$$

4.14 Service conditions.- The equipment shall be subjected to the environmental tests specified in Specification MIL-T-5422 with the following exceptions:

4.14.1 Temperature and altitude.- Replace the Temperature and Altitude tests as specified in MIL-T-5422 with the following:

4.14.1.1 Elevation.- The equipment, with fluid and film present and placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.11	Operation

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(b) Reduce the barometric pressure to 16.9 ± 0.1 inches of mercury (simulated 15,000 feet above sea level); maintain the temperature at +5° ± 1°C; and after temperature and pressure stabilization, repeat the measurement specified in 4.14.1(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level) and maintain the temperature at +38° ± 1° C. Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.14.1.1(a). The equipment shall meet specified performance for those measurements.

4.14.1.2 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 except the temperature at Step 4 shall be +100° F and at Step 9 shall be +41°F. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1, 4, 5, 9, and 10	4.11	Operation	Full specification requirements

4.14.2 Bounce and shock.- In addition to the shock test specified in Specification MIL-T-5422, the equipment shall be subjected to the following tests:

4.14.2.1 Bounce test.- The equipment, in its transit case, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The equipment shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 284 ± 2 rpm, for a total of 3 hours. During each 1/2 hour period of the test, the case shall rest on a different face. After the test, the equipment shall meet full specification performance requirements. There shall be no physical damage to the equipment or transit case. Minor surface abrasion to the transit case will be permissible.

4.14.2.2 Shock test; drop.- The equipment, in its transit case, shall be subjected to eight (8) thirty-inch drops. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame. Drops shall be made from a quick-release hook; or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. As a result of the test, the equipment shall not be impaired either mechanically or electrically. There shall be no degradation in equipment performance. Minor mechanical damage to the transit case will be permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.15 Standby operation.- The equipment shall be placed in operation and operated for 1/2 hour. The equipment shall then be stopped. After 4 hours, the equipment shall be started again and operated for another 1/2 hour. The performance shall be as specified in 3.12.

4.16 Suppression.- The equipment shall be tested in accordance with Specification MIL-I-6181 to determine compliance with the requirement specified in 3.14.

4.17 Visual and mechanical.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table V.

Table V.- Classification of visual and mechanical defects

Classification	Defect
Major	1. Illuminator housing cannot be easily released and swung open. 2. Access doors do not fit properly and cannot be opened and closed without difficulty. 3. Removable parts, such as take-up spool, cassette, supply tank, etc cannot be removed without use of tools. 4. Data chamber and card cannot be easily removed and replaced. 5. Guide rollers do not rotate freely. 6. Scratches on optical surfaces. 7. Controls not functioning properly.
Minor	1. Screws and fasteners missing. 2. Finish not as specified. 3. Controls do not operate smoothly and freely.

4.18 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP shall be preserved, packaged and tested in accordance with the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration and damage during worldwide shipment, handling and open storage.

5.1.2 Level C.- Recorder-Processor-Viewer, Radar-Mapping, RO-166( )/UP shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice.

5.2 Packing.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown unless an individually packed item exceeds that amount. Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, 1/2 inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping.

5.2.1 Level A.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class 2)	PPP-B-636	Table VI
Nailed Wood (Style 4, Class 2)	PPP-B-621	200 lbs
Wirebound (Style 2 or 3, Class 2)	PPP-B-585	200 lbs
Wood Cleated Fiberboard (Overseas)	PPP-B-591	200 lbs
Wood Cleated Plywood (Overseas)	PPP-B-601	200 lbs
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs

5.2.2 Level B.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class 1)	PPP-B-636	Table I
Nailed Wood (Style 4, Class 1)	PPP-B-621	200 lbs
Wirebound (Style 3, Class 1)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Domestic)	PPP-B-591	200 lbs
Wood Cleated Plywood (Domestic)	PPP-B-601	200 lbs
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5.2.3 Level C.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the modeiof transportation.

5.2.4 Pilot Pack.- When Level "A" packaging and packing is specified above, one acceptable model of Recorder-Processor-Viewer, Radar Mapping RO-166( )/UP will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustrations of the packaging and packing procedure, ' aether with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-262100.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Recorder-Processor-Viewer, Radar Mapping, RO-166( )/UP is used as a rapid processing strip film recorder for the continuous recording, processing, and viewing of side-looking radar reconnaissance information.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) Preproduction inspection:

Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(d)Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.16.)

(g) Quantity of tools and running spare parts required. (See 3.16.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

6.3 Nomenclature.- The parenthese in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RO-166W/UP. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Project 5841-A001

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.